

Project 33-N: Preliminary Multidisciplinary Design and Embedded Trigger-Timing Evaluation of a COTS-Based Guided Research Platform

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Abstract—This paper presents a preliminary multidisciplinary redesign of a commercial-off-the-shelf (COTS)-based guided research platform incorporating proposed folding-fin kinematics, deterministic embedded control, edge computation, and radiological sensing. The study combines analytical mechanism synthesis, preliminary structural, aerodynamic, and steady-state thermal modeling, an edge-inference latency benchmark, and source-reported electrical trigger-timing measurements. A four-bar linkage is proposed for fin deployment, with a target deployment time of 15 ms and a calculated spring-rate target of at least 0.185 N-m/rad. Under the stated assumptions, the hinge-pin model predicts a factor of safety of 1.88, while the aerodynamic model places the nominal static margin within the selected preliminary range near low angle of attack. Across 500 GPIO transitions, the revised ESP32 high-resolution timer implementation reduced the reported mean timing error from 2.4 ms to 0.12 ms and the reported standard deviation from 0.9 ms to 0.03 ms. These measurements validate only the electrical trigger signal. The edge-compute benchmark required 1.85 ± 0.15 s for 16 response tokens and is therefore treated as stationary or ground-side. Mechanical deployment, dynamic stability, transient thermal behavior, powered flight, and isotope-classification performance remain to be validated.

Index Terms—Project 33-N, systems integration, folding-fin kinematics, embedded timing, edge computing, radiological sensing, verification and validation

I. INTRODUCTION

Project 33-N investigates whether a compact COTS-based research platform can combine mechanical deployment, deterministic embedded timing, edge computation, and stationary radiological-sensing analysis under severe mass, volume, and power constraints. The present paper is not a flight-readiness, certification, or operational-capability claim. Its purpose is to establish a traceable preliminary design record and to separate analytical, modeled, benchmarked, measured, proposed, and unvalidated results.

The public Project 33 repository describes a baseline platform using a three-dimensional-printed airframe, folding tail fins, and an ESP32-based flight computer [1]. Any guided-flight result associated with that repository is treated here as repository-reported background and was not independently evaluated in this study. The proposed Project 33-N architecture adds a scintillation-sensing concept and a separate edge-compute node while intending to retain deterministic command authority on the microcontroller.

TABLE I
EVIDENCE-STATE DEFINITIONS

State	Meaning in this paper
ANALYTICAL	Derived from equations, assumptions, or design synthesis.
MODELED	Produced by simulation under stated assumptions.
BENCHMARKED	Measured for a computing or laboratory condition, not the integrated system.
MEASURED	Observed for a defined electrical or physical subsystem.
PROPOSED	Intended architecture or capability not yet demonstrated.
UNVALIDATED	No supporting test result is supplied.
ILLUSTRATIVE	Conceptual reconstruction, not native engineering evidence.

The baseline design record identifies three principal limitations: 1) thermal-structural vulnerability associated with polylactic-acid components near the propulsion section; 2) asymmetric electrical triggering associated with software scheduling delay; and 3) limited aerodynamic restoring behavior from flat-plate fins. The redesign addresses these issues through mechanism synthesis, material substitution, preliminary multiphysics modeling, and high-resolution timer dispatch.

II. EVIDENCE-STATE FRAMEWORK

To make claim maturity explicit, this paper uses the evidence states in Table I.

The most defensible present claim is narrow: Project 33-N establishes a preliminary multidisciplinary design baseline and a substantially improved source-reported electrical trigger-timing result. It does not establish physical fin deployment, flight performance, radiological classification, autonomous engagement, or integrated operational capability.

III. RELATED WORK

Folding-fin mechanisms may use torsion springs or pyrotechnic actuation and can experience rebound or asymmetric deployment [2]. Four-bar linkage synthesis using the Freudenstein equation provides a deterministic design alternative [3], [7]. In constrained edge computing, QLoRA enables parameter-efficient model adaptation [9], while low-bit quantization reduces memory demand. The contribution here is not a new theory within any single discipline; it is a preliminary evidence-controlled systems-integration study.

IV. ANALYTICAL MECHANISM DESIGN

A. Four-Bar Linkage Synthesis

ANALYTICAL/PROPOSED: A planar four-bar linkage was synthesized to move the fin from $\theta_2 = 0^\circ$ to $\theta_2 = 90^\circ$, with a target—not measured—deployment time of $t_{\text{deploy}} = 15$ ms and a nominal 92° over-center position. The Freudenstein relation is

$$K_1 \cos \theta_2 + K_2 \cos \theta_4 + K_3 = \cos(\theta_2 - \theta_4), \quad (1)$$

where $K_1 = r_1/r_4$, $K_2 = r_1/r_2$, and

$$K_3 = \frac{r_1^2 + r_2^2 - r_3^2 + r_4^2}{2r_1r_2}. \quad (2)$$

The controlling dimensions are $r_1 = 30$ mm, $r_2 = 10$ mm, $r_3 = 35$ mm, and $r_4 = 40$ mm. The nominal mid-stroke transmission angle is approximately 45° . Physical deployment time, friction, rebound, locking repeatability, and synchronization remain unvalidated.

B. Spring-Rate Target

ANALYTICAL: The spring-rate requirement is bounded by

$$\frac{1}{2}k\theta_{\text{max}}^2 \geq \frac{1}{2}I_{\text{fin}}\omega^2 + \int_0^{\theta_{\text{max}}} M_{\text{aero}}(\theta) d\theta + E_{\text{damping}}, \quad (3)$$

with the approximate aerodynamic hinge moment

$$M_{\text{aero}}(\theta) \approx \frac{1}{2}\rho v^2 C_{m_\alpha} c S_{\text{fin}} \sin(2\theta). \quad (4)$$

For the assumed mass, geometry, velocity, and damping inputs, the calculation gives $k \geq 0.185$ N-m/rad. This is a design target, not an experimentally verified spring rate.

V. MODELED STRUCTURAL, AERODYNAMIC, AND THERMAL BEHAVIOR

A. Controlling Model Configuration

The manuscript adopts the following controlling preliminary configuration. The CFD mesh count is fixed at approximately 4.2 million cells; the previously displayed 6.2-million-cell value is not used in this revision.

- **FEA—MODELED:** Ansys Mechanical; equivalent 50 g axial hinge-pin load case; Ti-6Al-4V; nominal 0.5-mm element size; pin-end constraints; cylindrical-contact load transfer.
- **CFD—MODELED:** Ansys Fluent; steady three-dimensional RANS; $k-\omega$ SST; approximately 4.2 million polyhedral cells; target $y^+ < 1$; domain extending 10 calibers upstream and 20 downstream; Mach 0.25 (80 m/s) at sea level.
- **Thermal—MODELED:** Steady-state compartment model using the stated heat-load, ambient, convection, and material assumptions. Detector integration time, transient propulsion heating, interface conductance, hot-ambient cases, and temperature-dependent degradation are outside the present model.

TABLE II
PROPOSED MATERIAL SUBSTITUTIONS

Component	Material	Rationale
Airframe	Al 7075-T6	Higher strength than PLA
Fins	CFRP	Reduced aft mass
Hinge pins	Ti-6Al-4V	High specific strength
AI node	ARM SBC	Quantized inference host

B. Preliminary Results and Boundaries

MODELED: The hinge-pin model reports a peak von Mises stress of 468 MPa under the stated equivalent load case. Relative to an assumed Ti-6Al-4V yield strength of 880 MPa, the modeled factor of safety is 1.88. Native solver files, load derivation, mesh convergence, contact sensitivity, fatigue, impact, and temperature-dependent properties are not supplied.

MODELED: The proposed aft-mass reduction moves the modeled center of gravity forward by approximately 4 mm. Static margin is

$$SM = \frac{X_{CP} - X_{CG}}{d}. \quad (5)$$

The predicted static margin lies within the selected 1.5–2.0-caliber range near the nominal and low-angle-of-attack condition, but decreases with angle of attack. The result does not establish dynamic stability.

MODELED: The steady-state thermal model reports a maximum edge-compute-compartment temperature of approximately 83.7°C under a 6.5-W heat load and the selected boundary conditions. This does not establish transient propulsion-heating survival or physical thermal correlation.

VI. PROPOSED EDGE-COMPUTE AND RADIOLOGICAL-SENSING ARCHITECTURE

A. Authority Separation

PROPOSED/UNVALIDATED: A multichannel analyzer would perform energy binning and peak extraction before formatting features for the edge-compute node. The current concept uses an ESP32 for timing-critical state estimation and bounded control logic, and a Raspberry Pi 5 with 8 GB RAM for INT4-quantized GGUF inference through `llama.cpp`.

AI output is advisory only in the proposed architecture. It is not evidence of direct actuator authority. A complete implementation must define accepted inputs, permitted outputs, confidence handling, timeout behavior, an “unknown/insufficient evidence” path, deterministic acceptance logic, safe rejection, logging, and independent verification that AI output cannot directly command actuation. Any numerical confidence threshold shown in illustrative material is not calibrated and is not an operative value.

B. Inference Benchmark

BENCHMARKED: Across $N = 50$ trials, generation of 16 response tokens after a 64-token prompt required 1.85 ± 0.15 s, corresponding to approximately 8.6 tokens/s; model-loading time was excluded. Detector acquisition, preprocessing, photon-integration time, transport, decision logic, and actuation latency

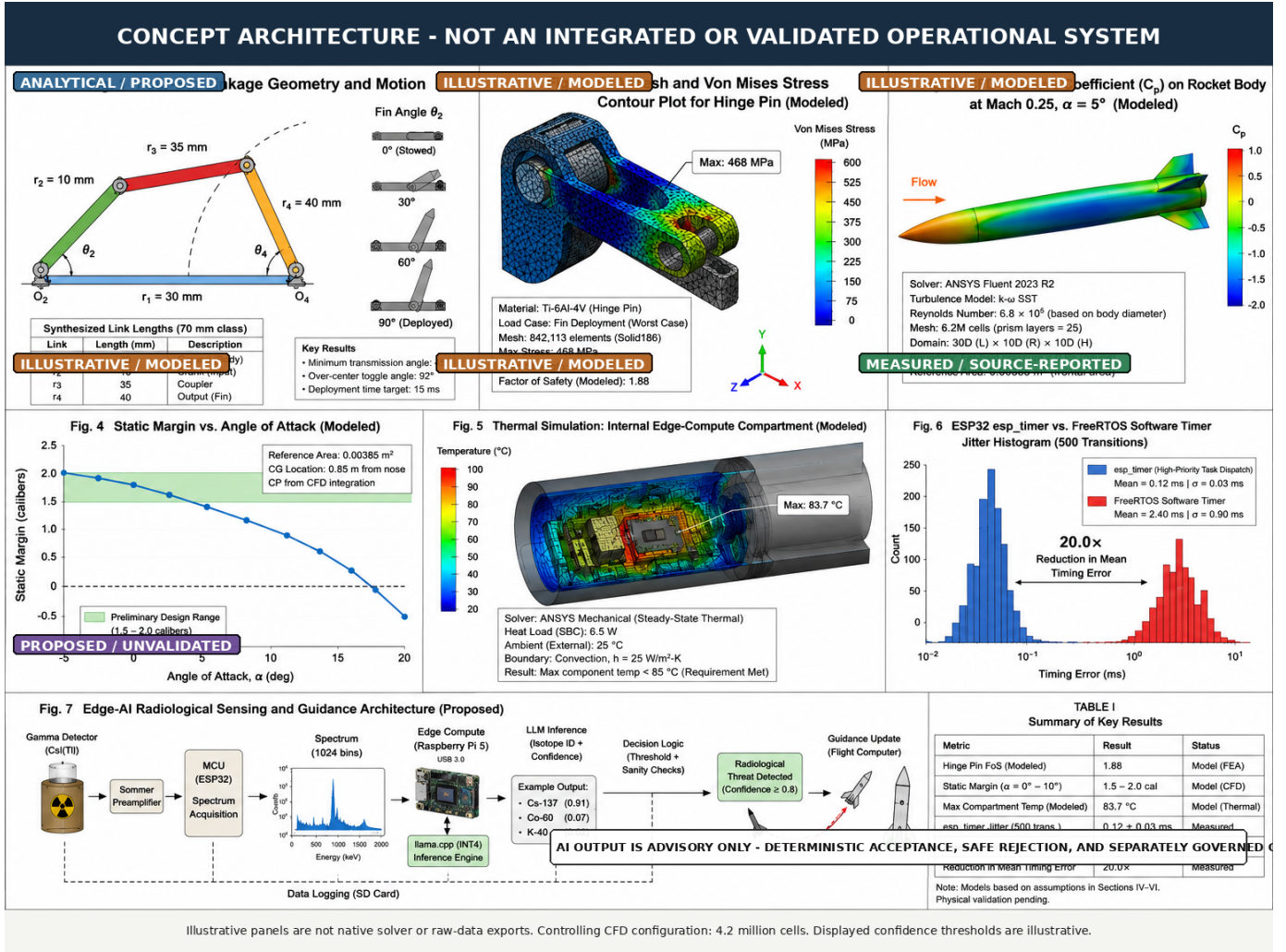


Fig. 1. Evidence-labeled overview. The mechanism panel is analytical/proposed; structural, aerodynamic, static-margin, and thermal panels are illustrative reconstructions of modeled claims rather than native solver exports; the timer panel summarizes source-reported measured electrical statistics rather than a plot generated from the raw dataset; and the sensing architecture is proposed and unvalidated. AI output is advisory within the proposed design and remains subject to deterministic acceptance, safe rejection, and separately governed command authority.

were not included. The result therefore supports only a computing benchmark and not end-to-end radiological identification. The present configuration is treated as stationary or ground-side.

VII. MEASURED ELECTRICAL TRIGGER TIMING

A. Timer Architecture

The baseline firmware used a synchronous `loop()` structure and software delays. The revised implementation uses the ESP-IDF `esp_timer` API [5], with callback dispatch through a high-priority FreeRTOS timer task rather than the ordinary application loop. This paper does not claim formal hard-real-time guarantees, zero latency, or an “instant” trigger.

B. Source-Reported Measurements

MEASURED/SOURCE-REPORTED: A logic analyzer was used for $N = 500$ GPIO transitions [1]. The baseline implementation reported a mean timing error of 2.4 ms with a standard deviation

Algorithm 1 High-Resolution One-Shot Trigger

Require: `trigger_sent = false`

- 1: Configure a one-shot high-resolution timer
- 2: Register `Timer_Deploy_Callback`
- 3: Start the timer using the prescribed delay
- 4: **function** `Timer_Deploy_Callback(arg)`
- 5: Set the electrical trigger output
- 6: `trigger_sent` $\leftarrow$ **true**
- 7: **end function**

of 0.9 ms. The revised implementation reported a mean timing error of 0.12 ms with a standard deviation of 0.03 ms. The ratios of the rounded values are 20.0 $\times$ for mean timing error and 30 $\times$ for reported standard deviation.

The raw dataset, logic-analyzer model, sampling configuration, timer settings, firmware commit, processor workload, tail-latency statistics, and independent reproduction were not supplied with the reviewed package. Consequently, this result

TABLE III
COMPACT CLAIM-TO-EVIDENCE MATRIX

Area	State	Highest-priority missing evidence
Linkage	Analytical / proposed	Deployment time, friction, rebound, locking, synchronization
Hinge pin	Modeled	Native model, convergence, fatigue, impact, thermal effects
Aerodynamics	Modeled	Dynamic derivatives, wind-tunnel correlation, flight data
Thermal	Modeled	Transient heating and physical correlation
Edge inference	Benchmarked	End-to-end latency, calibration, classifier comparison
Trigger timing	Measured*	Raw data, full test configuration, tail latency, reproduction
Radiological ID	Proposed	Calibrated spectra, accuracy, false alarms, unknown handling
Integrated system	Unvalidated	End-to-end authority, safety, sensing, control, and flight evidence

*Source-reported electrical measurement.

validates only the source-reported electrical trigger signal and does not establish physical deployment time, fin synchronization, rebound, locking, or flight performance.

VIII. CLAIM-TO-EVIDENCE MATRIX

IX. REPRODUCIBILITY, AVAILABILITY, AND SCOPE

A. Configuration and Reproducibility

The controlling configuration for this revision is the one stated in the manuscript: 4.2 million CFD cells, the listed 70-mm-class linkage dimensions, the equivalent 50 g hinge-pin load case, and the source-reported timing statistics. Future releases should identify exact geometry revisions, solver projects, mesh versions, firmware commits, hardware revisions, prompt templates, quantization settings, sampling parameters, and statistical scripts.

B. Data and Code Availability

The Project 33 repository is cited in [1], and the referenced model record is cited in [8]. The raw 500-transition timing dataset, native FEA/CFD/thermal projects, convergence histories, detector datasets, and end-to-end test records were not supplied with this revision and are therefore not available for independent reproduction through this manuscript.

C. Funding and Conflict of Interest

No external funding is reported for this study. The author is the founder of Vers3Dynamics, the organization associated with the platform and software described in this paper.

D. Safety and Dual-Use Scope

This paper documents preliminary research and does not establish operational readiness, autonomous engagement, flight certification, safety approval, or legal authorization. The described radiological-sensing architecture is proposed and unvalidated. Any physical testing or broader technical disclosure should be conducted only under applicable legal, safety, range, regulatory, and export-review requirements.

X. LIMITATIONS AND FUTURE WORK

This work does not provide native solver exports, convergence evidence, transient thermal validation, physical mechanism measurements, powered-flight data, dynamic-stability measurements, calibrated radiological datasets, isotope-classification accuracy, or end-to-end system validation. The next credibility step is a stronger evidence chain rather than a broader claim: release a reconciled configuration record; publish native solver outputs and convergence evidence; provide the raw timing data and reproducible plotting code; test mechanism deployment, rebound, locking, and synchronization; evaluate calibrated stationary radiological sensing against conventional classifiers; and verify the advisory AI authority boundary in hardware and software.

XI. CONCLUSION

Project 33-N currently supports a preliminary multidisciplinary design baseline, a computing-latency benchmark, and one narrow source-reported electrical timing result. Under the stated assumptions, the models report a hinge-pin factor of safety of 1.88, a nominal low-angle static-margin result, and an 83.7°C steady-state compartment temperature. Across 500 transitions, the reported mean electrical timing error decreased from 2.4 ms to 0.12 ms and the reported standard deviation decreased from 0.9 ms to 0.03 ms. These findings do not establish physical deployment, flight performance, radiological classification, autonomous engagement, or integrated operational capability. Future releases should let traceable, reproducible evidence—rather than presentation intensity—carry the credibility of the project.

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